Quantum Phase Transitions of the Transverse-Field Ising Model

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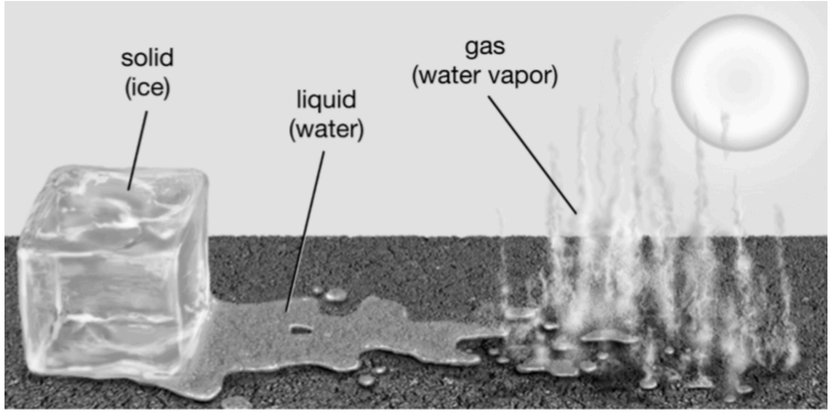
Phase transitions and the classical Ising model

- Phase transition = non-analyticity of a thermodynamic potential; symmetry-breaking $\to$ an **order parameter** zero in the disordered phase, nonzero (and non-unique) in the ordered one

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \quad \sigma_i = \pm 1$$

- $\mathbb{Z}_2$ symmetry: invariant under the global flip $\sigma_i \to -\sigma_i$; order parameter = magnetization m



First-order: ice $\to$ water $\to$ vapor

dim.	transition?	why
1D	**no**	$\Delta F_{\text{wall}} = 2J - k_B T \ln N < 0$ for any $T > 0$
2D	yes (Onsager '44)	wall boundary is extensive — entropy cost is payable

The transverse-field Ising model

- QPT: transition at $T = 0$ driven by quantum (not thermal) fluctuations — the **ground-state energy** becomes non-analytic in a coupling g
- Promote the Ising spins to Pauli operators and add a transverse field σ^x that does **not** commute with σ^z $\to$ quantum dynamics (tunnelling between classical configurations)

$$H = J \sum_{\langle i,j \rangle} \sigma_i^z \sigma_j^z + h \sum_i \sigma_i^x, \quad J = -1, \quad \langle |M_z| \rangle = \langle | \sum_i \sigma_i^z | \rangle$$

regime	ground state	phase
$g = h/\lvert J\rvert \gg 1$	$\prod_i \lvert -\rangle_i$ ($\sigma^x = -1$)	paramagnetic
$g \ll 1$	$\prod_i \lvert \uparrow\rangle_i$ or $\prod_i \lvert \downarrow\rangle_i$	ferromagnetic

- Finite N: avoided crossing (smooth); true non-analyticity only at $N \to \infty$ — exact 1D critical point $g_c = 1$

Two exact routes to $g_c = 1$

route	mechanism	gives
Suzuki-Trotter	$Z = \mathrm{Tr}\big(e^{-\beta H}\big)$ sliced into M imaginary-time steps $\to$ **classical** 2D anisotropic Ising ($N \otimes M$); Onsager's exact $\sinh(2\beta J_x^c)\sinh\big(2\beta J_y^c\big) = 1$, then $M \to \infty$	non-analytic free energy — **sufficient**
Jordan-Wigner	spin $\to$ fermion ($\sigma_j^z = 2\hat{n}_j - 1$) + nonlocal string $\to$ H **quadratic** $\to$ Fourier + Bogoliubov; $\varepsilon(k) = 2\sqrt{1 + g^2 - 2g\cos k}$	vanishing gap at $k = 0$ — **necessary**

- Both are exact and non-perturbative — no fit, no series expansion; **zero shared machinery, same $g_c = 1$** (derivations in backup)
- JW also gives $E_0 = -\sum_{k>0} \varepsilon(k)$ in closed form — evaluated by `src/physics/jordan_wigner.py` as a numerical cross-check later

Entanglement entropy and the area law

- A correlation-based probe, **independent of any order parameter** — and it fixes the **universality class** (via c), not just the transition's location
- Bipartite pure state $\to$ Schmidt decomposition; entanglement entropy $S = -\mathrm{Tr}(\rho_A \ln \rho_A)$ of the reduced density matrix
- Gapped ground states obey an **area law**: S scales with a region's boundary, not its volume — only correlations within a correlation length ξ of the cut matter
- At criticality $\xi \to \infty$: the area law breaks, S grows **logarithmically** with block size, coefficient fixed by the central charge c (Calabrese-Cardy) — Ising CFT: $c = 1/2$
- Exactly at $g \to 0$ the closed-form TFIM entropy predicts $S \to \ln 2$ — the cat-state degeneracy of Sec. "classical Ising model" made explicit

The dataset: qspin / Ising

- `qml.data.load("qspin", sysname="Ising", periodicity=.., lattice=.., layout=..)` — fully specified by four discrete choices: `periodicity` `"open"`/`"closed"`, `lattice` `"chain"` (1D)/`"rectangular"` (2D), `layout` = `rows x cols` (chain: `1x4`,`1x8`,`1x16`; rect.: `2x2 … 4x4`)
- $N = \mathrm{rows} \times \mathrm{cols} \in \{4, 8, 16\}$; each system sweeps 100 values of h, always straddling $h = 1$

attribute	shape	meaning
`hamiltonians`	$100 \times$ `qml.Hamiltonian`	$H(h)$, one per field — read directly by our code, not hard-coded
`ground_energies` / `ground_states`	$(100,)$ / 100×2^N	exact $E_0(h)$ and $\|\psi_0(h)\rangle$
`order_params`	$(100,)$	$\langle \|M_z\| \rangle(h)$
`shadow_basis` / `shadow_meas`	$(100, 1000, N)$	classical-shadow bases / outcomes

The code: one module per physics question

- Pipeline: cached `qspin` HDF5 $\to$ `loader.load_ising()` $\to$ typed `IsingData` $\to$ **one `src/*.py` per physics question** $\to$ one PNG in `plots/` $\to$ embedded by relative path into **both** `report.md` and these slides — **no number in either output is typed by hand**

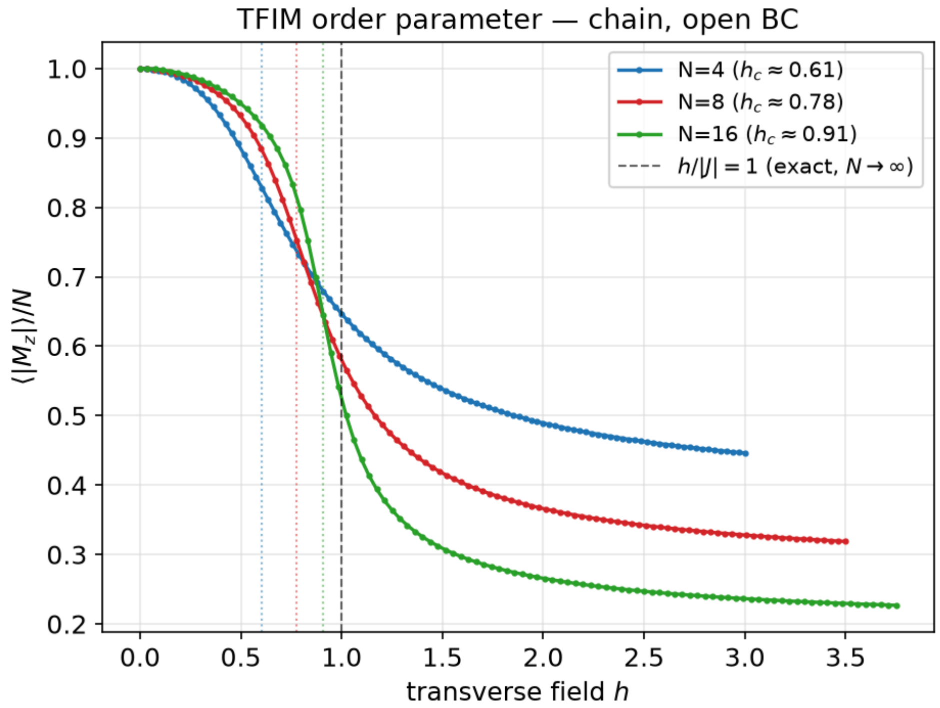
- `extract_structure` reads the dataset's own `qml.Hamiltonian` term-by-term (`PauliZ@PauliZ` $\to$ bonds, `PauliX` $\to$ field wires): **no hardcoded lattice** — the same VQE code is correct for open/closed, 1D/2D
- Shared `plotting.py` (one style, one $N \to$ colour map); `parallel.py` process pool for the multi-restart VQE grids

artefact	size
`src/` modules	16 (2 151 lines)
`tests/`	621 lines, **63 passing**
coverage	1 test module / physics module
figures	every PNG script-generated

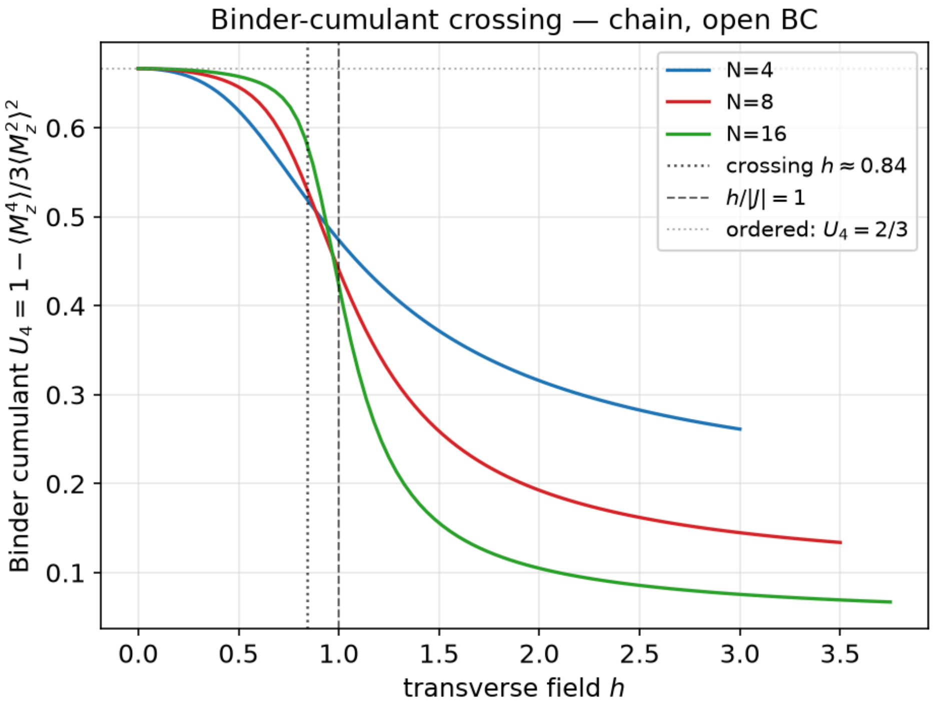
Exact results

Order parameter and the Binder cumulant

- Binder cumulant $U_4 = 1 - \langle M_z^4 \rangle / (3\langle M_z^2 \rangle^2)$ — cancels finite-size shape effects



$\langle |M_z| \rangle / N$ vs h

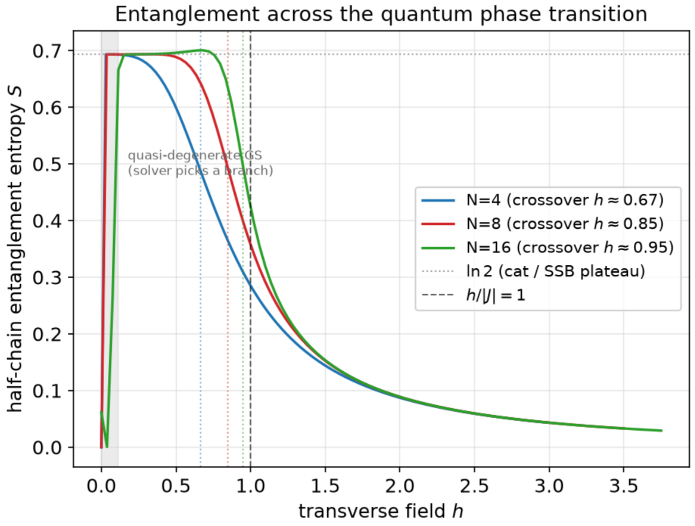


U_4 crossing ≈ 0.84

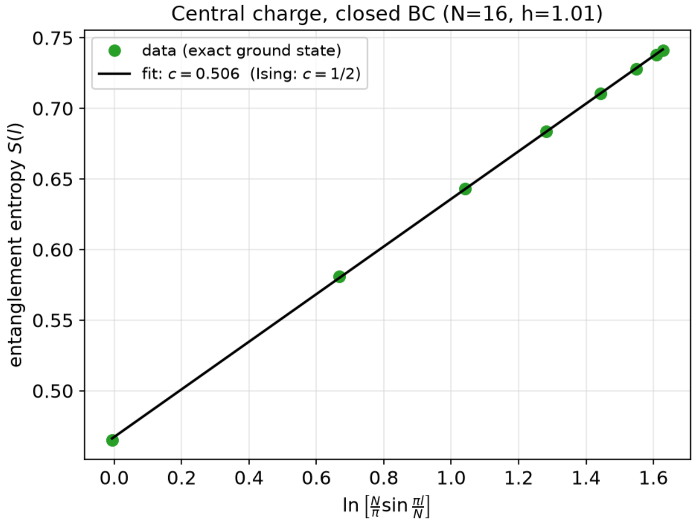
`src/physics/analysis.py`

Entanglement entropy and the central charge

- Three regimes: area-law plateau $\rightarrow$ critical peak ($\xi \rightarrow \infty$, a third independent g_c estimate) $\rightarrow$ $\ln 2$ plateau (ordered cat state)
- Calabrese-Cardy fit $S(l) = (c/6)\ln[(2N/\pi)\sin(\pi l/N)] + c_{1'}$: open chain $c \approx 0.62$ (boundary bias, arXiv:1002.4353) $\rightarrow$ **closed chain $c \approx 0.51$**, the Ising value — same code, boundary removed (backup)



S vs h: area law $\rightarrow$ peak $\rightarrow$ $\ln 2$



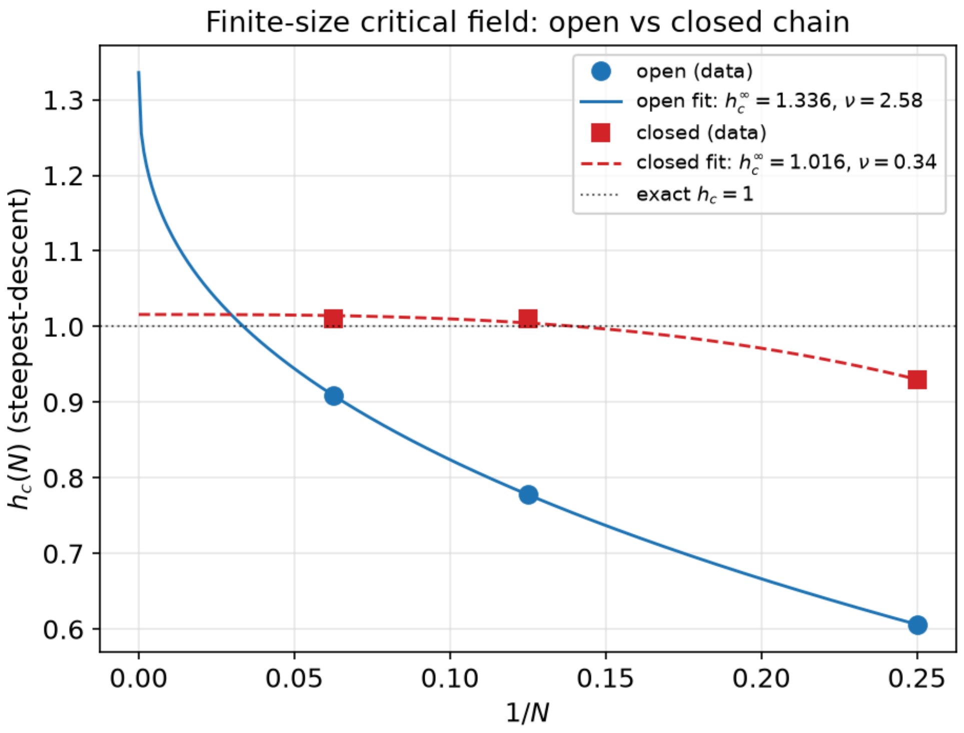
Closed chain: $c \approx 0.51$ (Ising: $1/2$)

`src/physics/entanglement.py`

Locating g_c: finite size and boundaries

N	h_c open	h_c closed
4	0.61	0.93
8	0.78	1.01
16	0.91	1.01

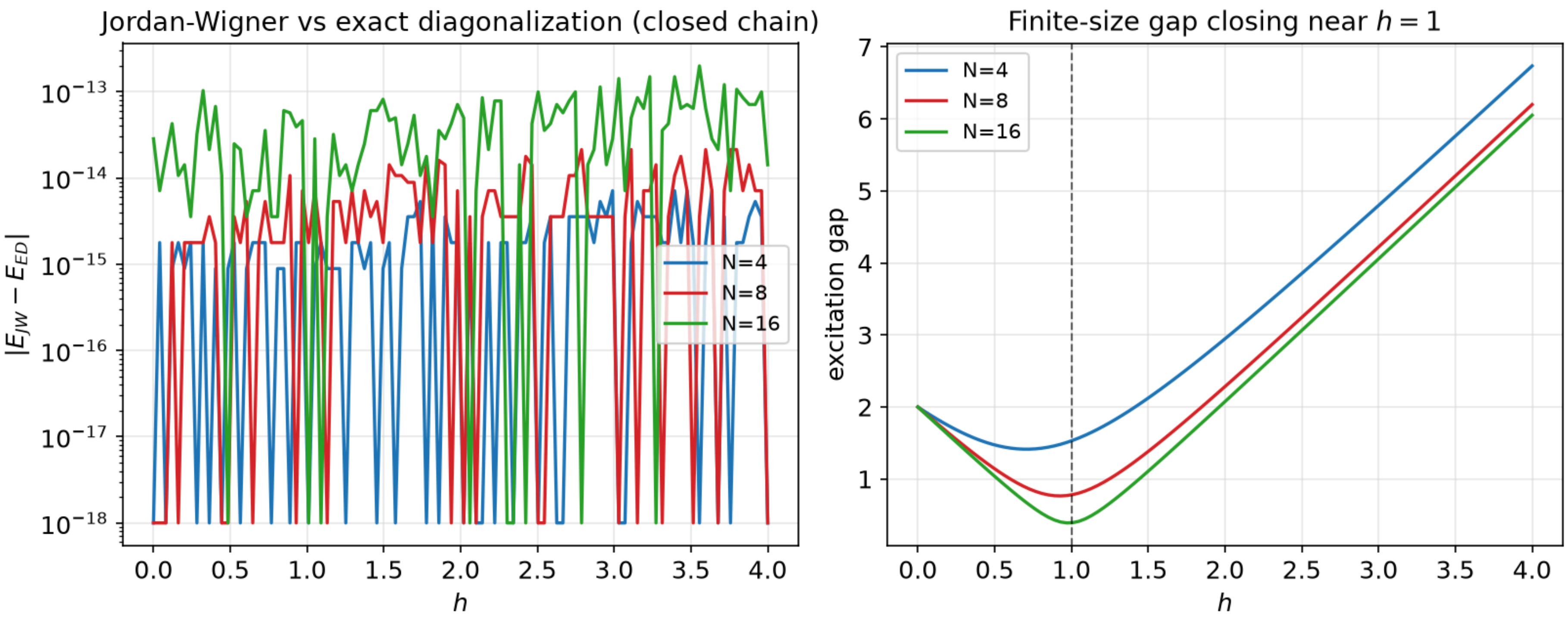
- Both drift toward $h/|J| = 1$ (**thermodynamic limit**); steepest-descent of $\langle |M_z| \rangle$ and the entropy crossover agree at each N
- Finite-size scaling $h_c(N) = h_c(\infty) - aN^{-1/\nu}$, $\nu = 1$ — the closed chain's a is far smaller: **no ends, no boundary correction**
- Same boundary physics that biased c — one cause, two symptoms



`src/physics/finite_size_scaling.py` · `src/physics/boundary_comparison.py`

Independent check: Jordan-Wigner

- The closed-form free-fermion energy (previous section) matches the dataset's exact diagonalization to floating-point precision, with no ED involved



`src/physics/jordan_wigner.py`

Variational quantum eigensolver

VQE: the variational principle and two ansätze

$$E(\theta) = \langle\psi(\theta)|H|\psi(\theta)\rangle \geq E_0, \quad |\psi(\theta)\rangle = U(\theta)|\psi_0\rangle$$

- Hybrid loop: a quantum device evaluates $E(\theta)$ and its gradient (adjoint), a classical optimizer (Adam) updates θ. Caveat: convergence certifies a **local** optimum of a **bounded** ansatz — $E(\theta^*) - E_0$ has an expressibility floor unless $|\psi_0\rangle$ lies in $U(\theta)$'s reachable manifold

ansatz	circuit	knows about H?
HEA	generic single-qubit rotations + entangling gates	no — expressive
HVA	p layers of $\prod_{\langle i,j\rangle} e^{-i\gamma_l Z_i Z_j} \prod_i e^{-i\beta_l X_i}$ — 2 params/layer	yes — reads H's bonds

- HVA reference state $|-\rangle^{\otimes N}$ = the **exact** ground state at $g \to \infty$; **warm-start** sweeps h high→low, each field initialized from the previous optimum — transporting the easy solution into the entangled regime

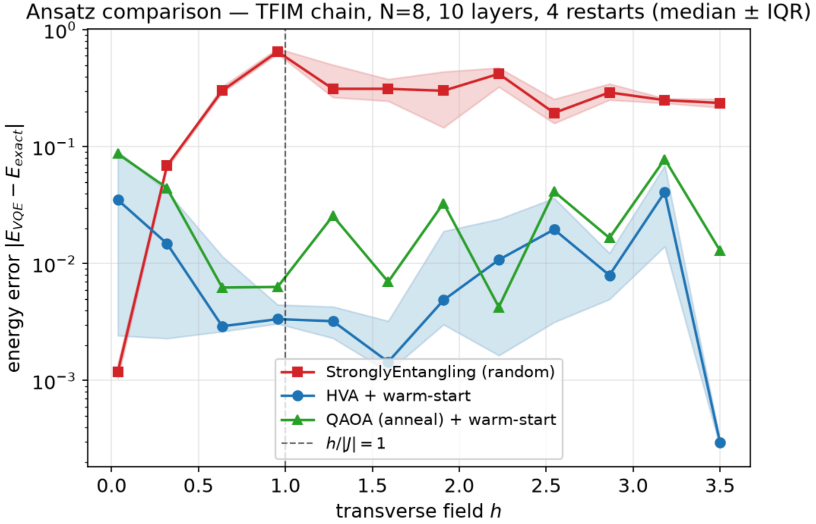
QAOA as digitized quantum annealing

- Adiabatic path from an easy Hamiltonian to H itself: $H(s) = (1 - s)H_B + sH_C$, $s : 0 \to 1$ slowly — the adiabatic theorem guarantees the ground state is tracked if s varies slowly enough relative to the gap
- One Trotter step of this sweep **is** one HVA layer: $H_B = \sum_i X_i$ (mixer), H_C = the ZZ coupling — the same (γ, β) circuit, reinterpreted
- **Annealing-inspired init**: layer l seeded at $s_l = (l + 1/2)/p$ with $\gamma_l = s_l \cdot \delta t$, $\beta_l = (1 - s_l) \cdot \delta t$ — places the optimizer in the basin a slow anneal would follow, instead of starting at random
- As $p \to \infty$ with a slow schedule, QAOA reproduces exact adiabatic evolution; at finite p the residual energy error measures the cost of digitization
- Same circuit as HVA $\to$ **same trainability**: the barren-plateau table reports one shared HVA/ QAOA column

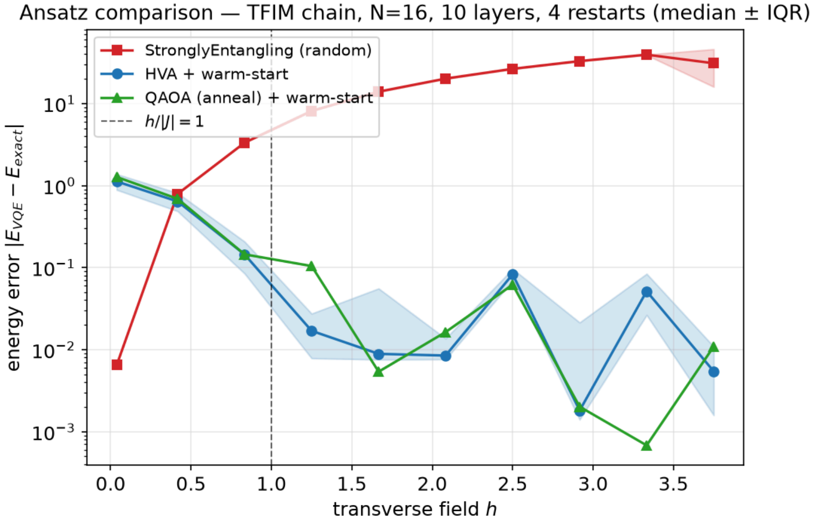
`src/ansatz/vqe_hva.py`

VQE: hardware-efficient vs physics-informed — results

- Error metric: $\Delta E(\theta) = E(\theta) - E_0 \geq 0$; each curve is the **median over 4 restarts**, band = IQR
- The HEA→HVA gap **widens** with size: $\approx 23\times$ at $N = 8 \to \approx 85\times$ at $N = 16$, where HEA hits its barren plateau



$N = 8$: median $|\Delta E|$ 2.8×10^{-1} (HEA) $\to 1.2 \times 10^{-2}$ (HVA), $\approx 23\times$



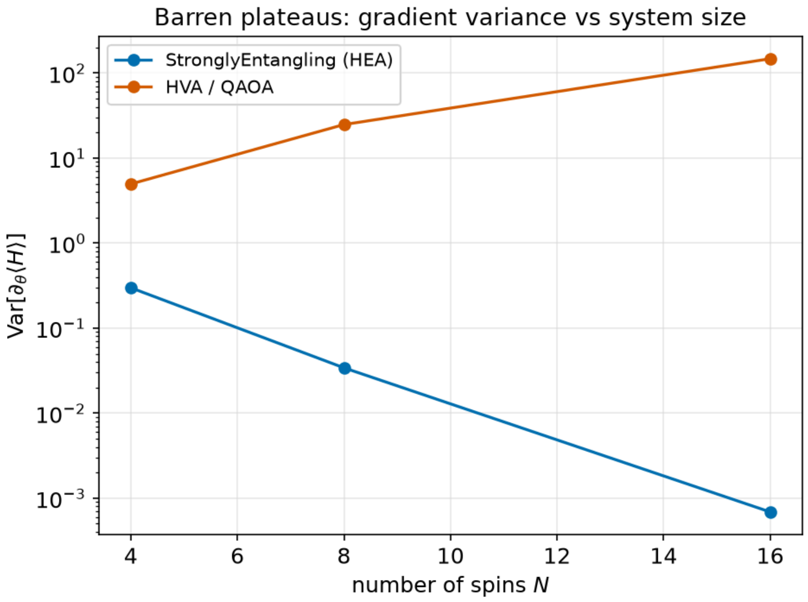
$N = 16$: 1.8×10^{1} (HEA) $\to 2.1 \times 10^{-1}$ (HVA), $\approx 85\times$

`src/ansatz/vqe.py` · `src/ansatz/vqe_hva.py` · `src/ansatz/benchmark.py`

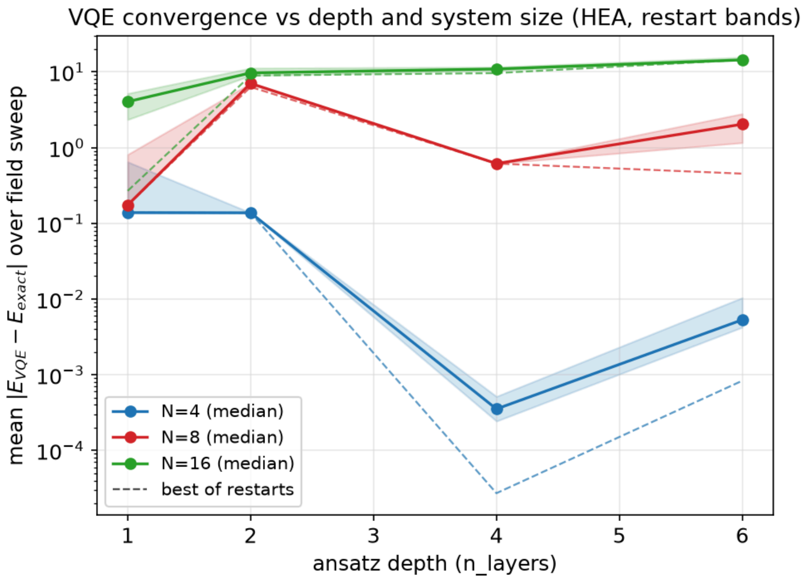
- Diagnostic: $\mathrm{Var}_\theta\left[\partial_{\theta_i}\langle H\rangle\right]$, averaged over components i across n random θ draws (protocol in backup). HEA collapses with N; HVA/QAOA **grows** — a polynomial-dimension dynamical Lie algebra (Larocca et al.), enforced by $\mathbb{Z}_2$ symmetry

N	HEA	HVA
4	0.298	5.0
8	0.034	24.8
16	$6.9 \cdot 10^{-4}$	147.3

`src/ansatz/trainability.py`
`src/ansatz/benchmark.py`



Gradient variance vs N



HEA depth scan: deeper = **worse** at $N = 16$

Summary and next steps

- Reproduced the TFIM phase transition from exact data, an independent analytic (Jordan-Wigner) solution, and VQE — all agree
- Physics-informed ansatz (HVA) beats generic HEA on accuracy and trainability
- Periodic boundaries resolve two open-chain artifacts at once: the central-charge bias and the slow h_c convergence
- Extended scope (backup): 2D lattices (h_c grows with connectivity) and classical-shadow reconstruction (the NISQ-realistic result)
- Open: combined order-parameter + entropy figure; error bars on the finite-size fits (only 3 system sizes)

Thank you

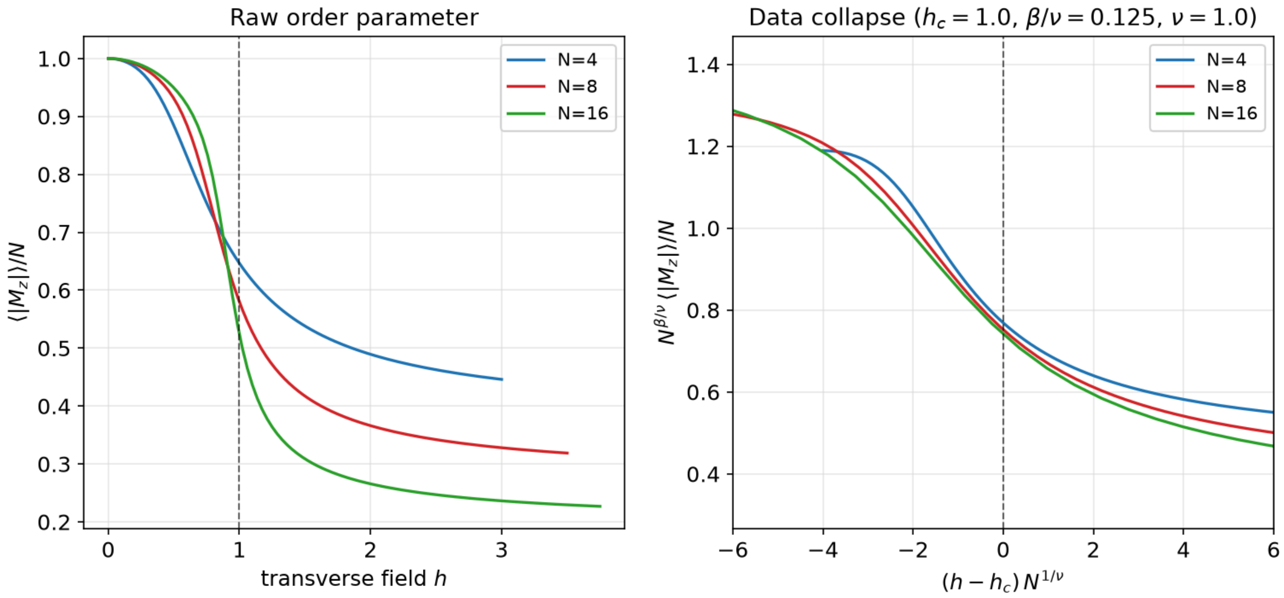
Backup slides

Backup: Suzuki-Trotter, the mapping in full

- Suzuki-Trotter: slice $Z = \mathrm{Tr}\left(e^{-\beta H}\right)$ into M imaginary-time steps $\to$ partition function of a **classical** 2D anisotropic Ising model, N sites $\otimes$ M time-slices — imaginary time **is** the second spatial dimension
- Quantum coupling g becomes a classical anisotropy $\beta^* J_y = \gamma = -\frac{1}{2}\ln[\tanh(\beta J g/M)]$
- Onsager's exact 2D critical point, $\sinh(2\beta J_x^c)\sinh\left(2\beta J_y^c\right) = 1$, substituted back with $M \to \infty$ (i.e. $T \to 0$) gives $g_c = 1$ for **every** J
- Exact and non-perturbative: no series expansion, no fit — and it gives the non-analyticity of the free energy itself (**sufficient** for a QPT), where JW's vanishing gap is only **necessary**

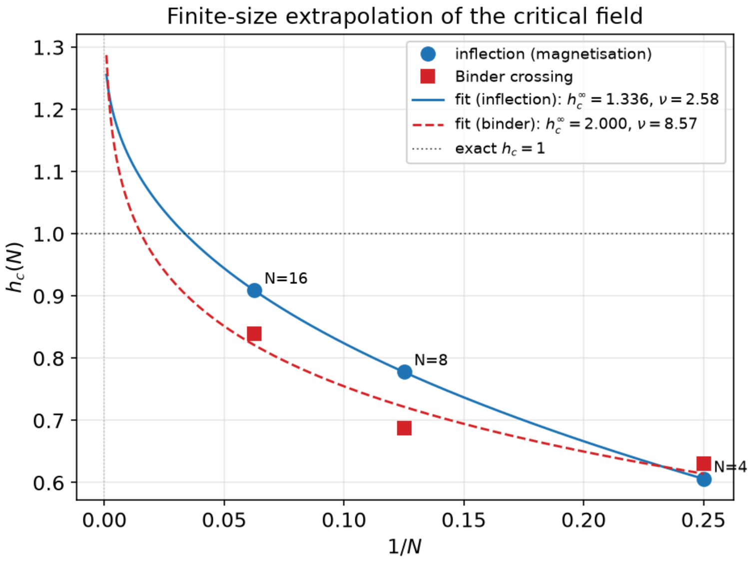
Backup: data collapse

- Data collapse: $(\langle|M_z|\rangle/N)N^{\beta/\nu} = f\left[(h-h_c)N^{1/\nu}\right]$ – collapses **only** for the correct h_c **and** exponents, so it fixes location + universality class at once (no central charge needed)
- $\beta = 1/8$, $\nu = 1$ are the 2D-Ising exponents (the imaginary-time dimension of the Suzuki-Trotter mapping)

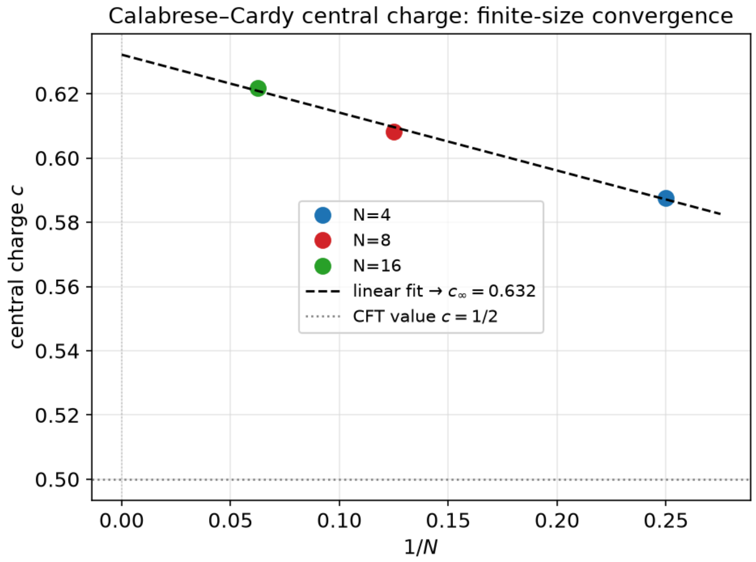


Backup: why we don't quote an $h_c(\infty)$

- We **measure** $h_c(N)$ and deliberately do **not** quote an extrapolated $h_c(\infty)$ — this slide is why
- Free fit of $h_c(N) = h_c(\infty) - aN^{-1/\nu}$ to 3 points: $h_c(\infty) = 1.34$ (inflection) / 2.00 (Binder — **its fit bound**), $\nu = 2.58/8.57$ vs exact $\nu = 1$; `curve_fit` cannot even estimate a covariance — 3 points, 3 free parameters
- Main deck therefore fixes $\nu = 1$ and quotes only measured $h_c(N)$; more sizes would need $N > 16$



Free 3-point fit overshoots $h_c = 1$

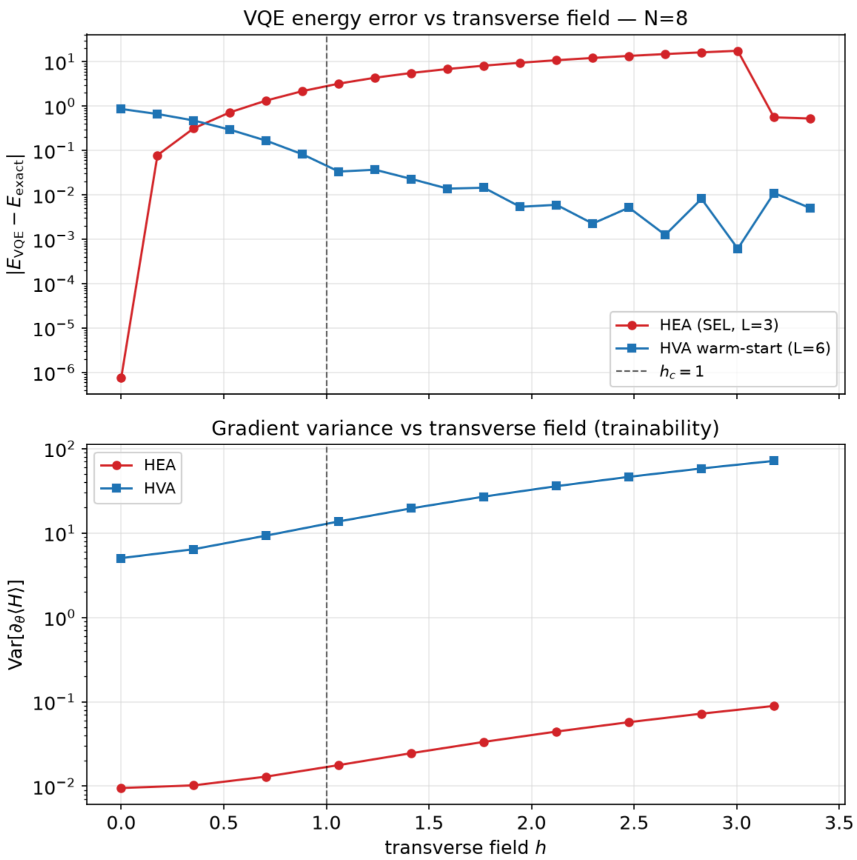


c drifts **away** from $1/2$ — boundary bias

src/physics/finite_size_scaling.py

Backup: where HVA's advantage actually lives

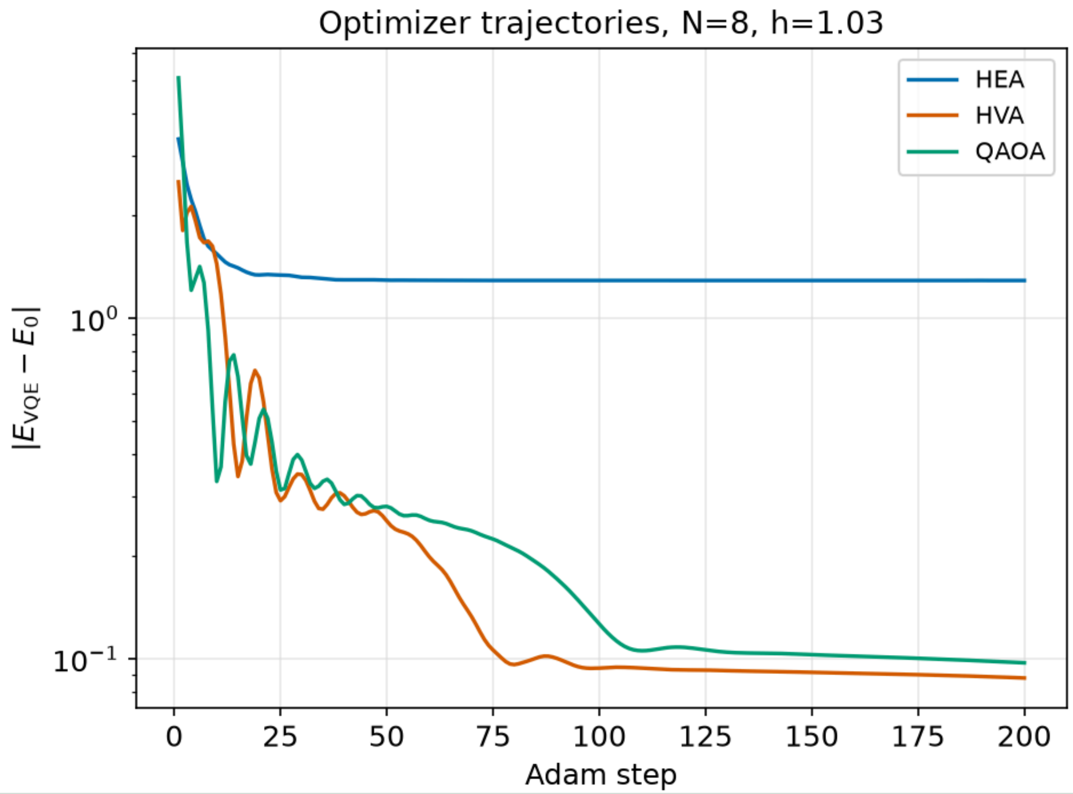
- The headline "$\approx 23\times$ at $N=8$" is a **median over the sweep** — resolved by field, the advantage is **not** uniform
- $h=0$: HEA is **better** by orders of magnitude (exact product state — why we exclude $h=0$); crossover at $h\approx 0.4$; HVA then wins by up to 10^4
- HVA's worst point is the deep ordered phase: its warm start is exact at $h\to\infty$, and a GHZ cat state needs depth $\sim N/2$
- Gradient variance (bottom): HVA above HEA at **every** field — the trainability gap is **not** a critical-point artifact; mid-sweep values reproduce the main-deck $N=8$ row from a **different** module



src/ansatz/vqe_error_profile.py

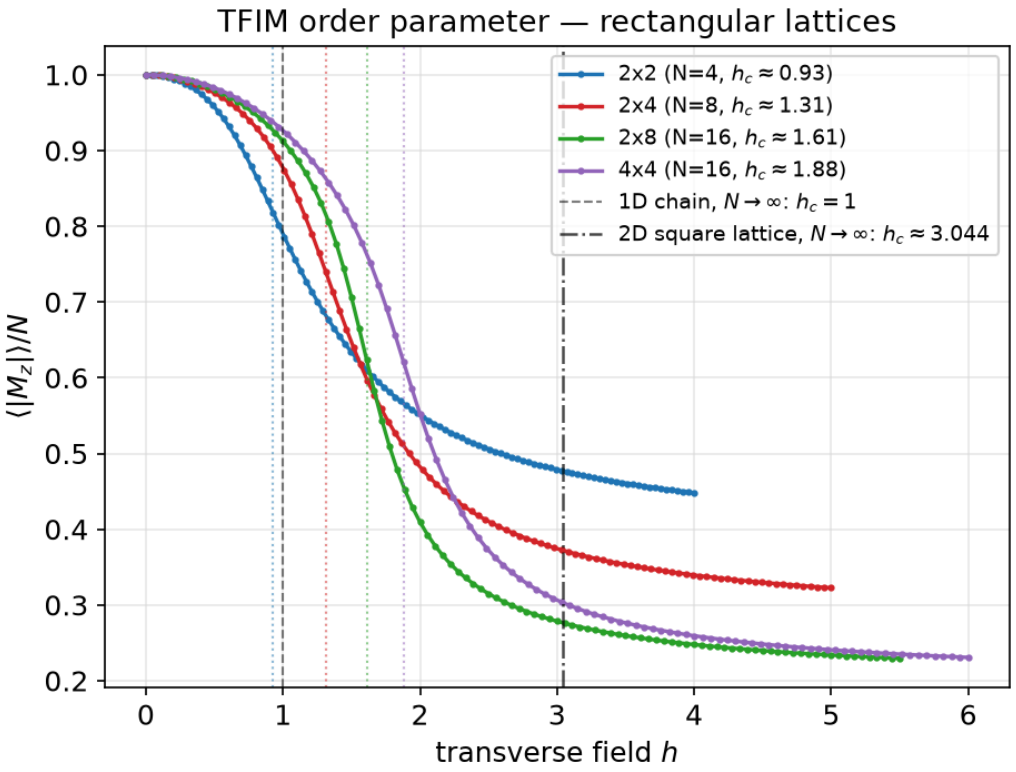
Backup: why HVA wins — optimizer trajectories

- ΔE vs Adam step at one field: HEA plateaus early, orders of magnitude above where HVA/QAOA end up — the endpoint comparison alone **understates** the gap

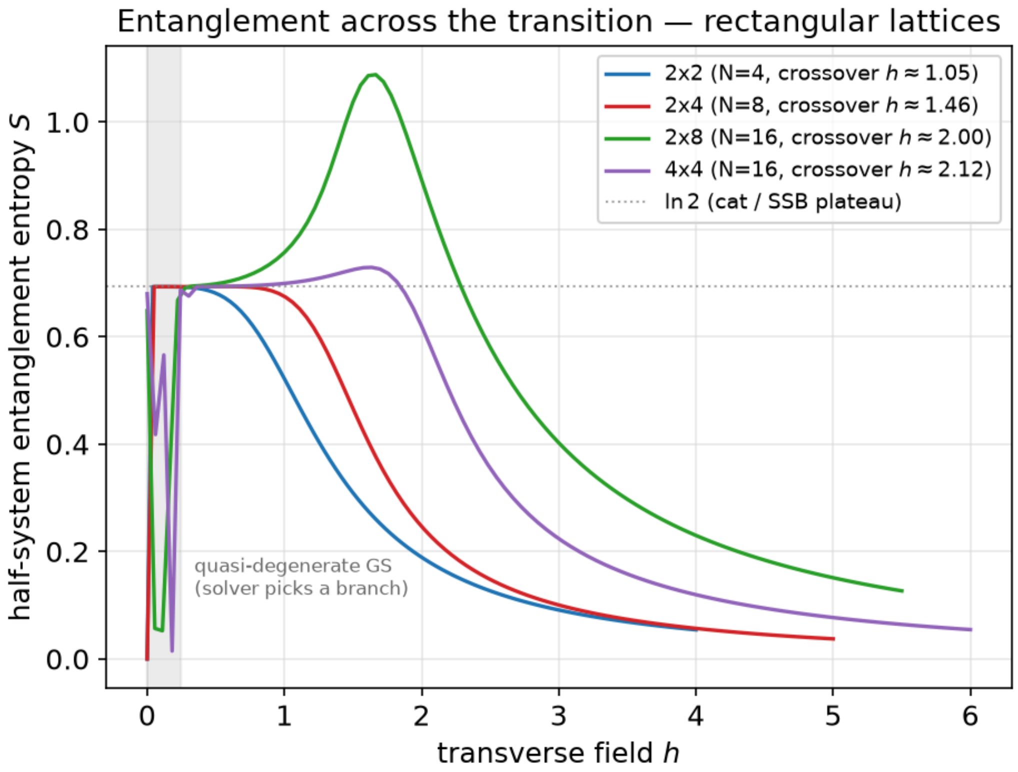


Beyond the original scope

Backup: 2D lattices



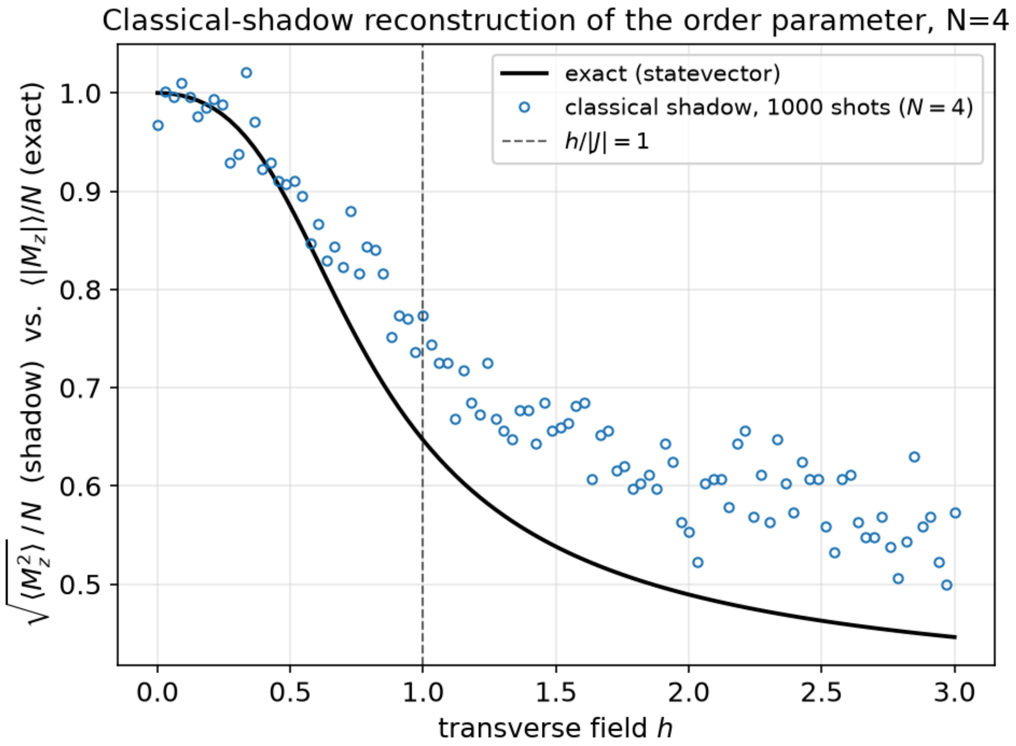
Order parameter, 2x2/2x4/2x8/4x4



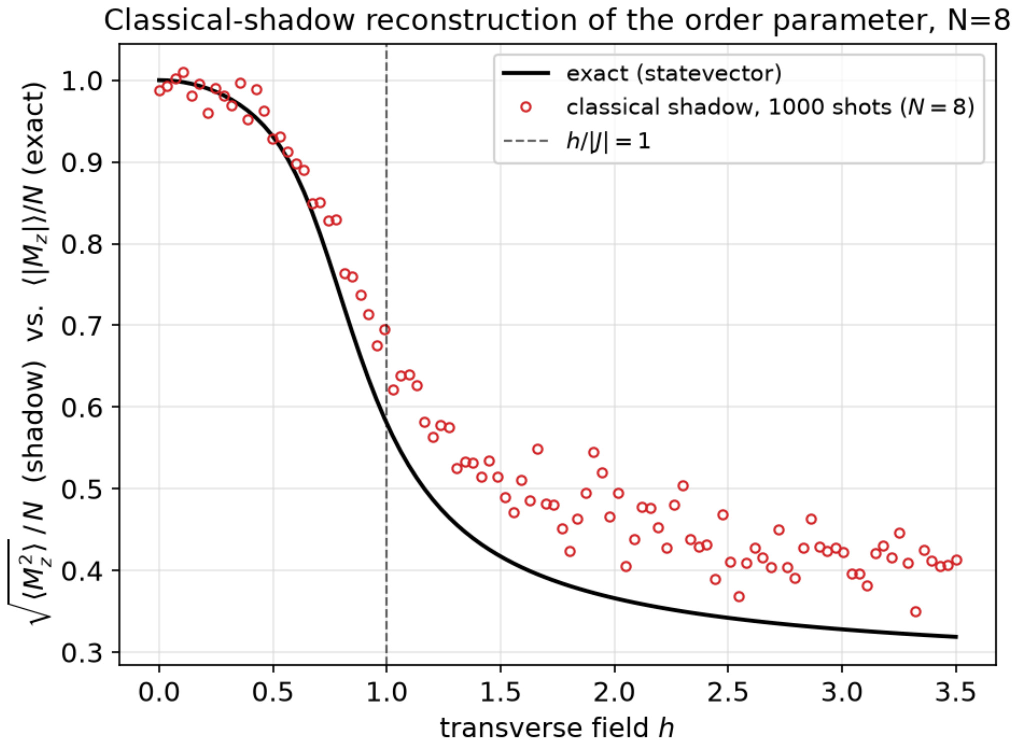
Entanglement entropy, 2x2/2x4/2x8/4x4

Backup: classical-shadow reconstruction

- Shadow estimator: $\hat{\rho} = \otimes_i \, (3|b_i\rangle\langle b_i| - \mathbb{I})$ per snapshot, averaged (median-of-means) over 1000 shots



$N = 4$



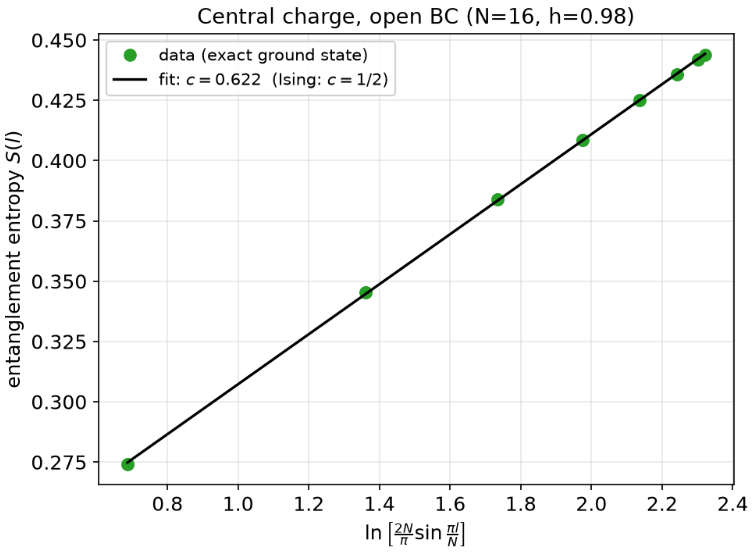
$N = 8$

Backup: why $\sqrt{\langle M_z^2 \rangle}$?

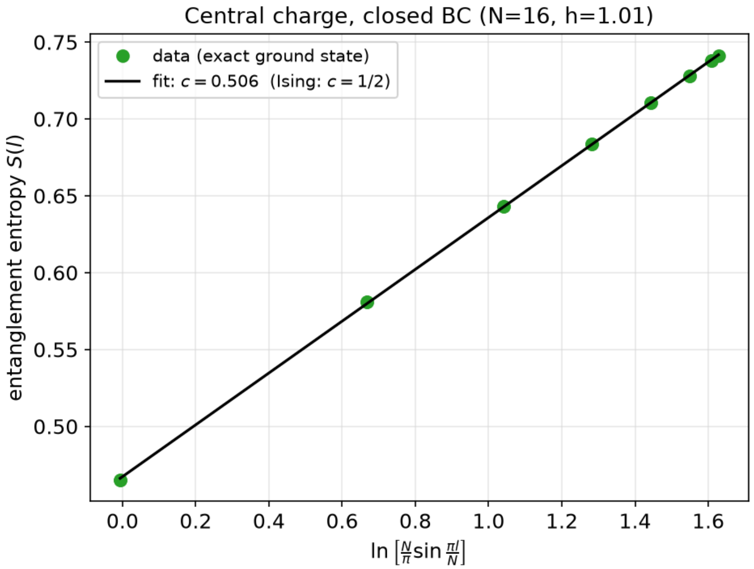
- Classical shadows give unbiased estimators of **linear** functionals of ρ only: $\langle O \rangle = \mathrm{Tr}(\rho O)$ for an operator O
- $\langle |M_z| \rangle$ is **not** linear in ρ (absolute value), so it cannot be estimated this way — no shadow protocol gets around this
- $M_z^2 = \sum_i Z_i^2 + \sum_{i \neq j} Z_i Z_j$, so $\langle M_z^2 \rangle = N + \sum_{i \neq j} \langle Z_i Z_j \rangle$ **is** linear — a genuine two-body Pauli observable
- If M_z only ever takes values $\pm m$ (deep ordered phase), $\langle |M_z| \rangle = m = \sqrt{\langle M_z^2 \rangle}$ exactly; elsewhere it is an approximation, worst near criticality

Backup: fixing the central-charge bias

- Calabrese-Cardy (open chain): $S(l) = (c/6)\ln[(2N/\pi)\sin(\pi l/N)] + c_{1'}$ — the naive 2-parameter fit used throughout the report
- Real correction (arXiv:1002.4353): boundary-localized **irrelevant operators** add subleading terms scaling as a power of N, biggest at the two open ends. Lesson: a persistent bias against known theory is itself data — chase the physics before suspecting the code



Open: $c \approx 0.62$ — boundary present



Closed: $c \approx 0.51$ — bias gone

Backup: building the HVA circuit

- `extract_structure(hamiltonian)` inspects the dataset's own `qml.Hamiltonian` term-by-term: `PauliZ@PauliZ` factors $\to$ bonds, lone `PauliX` factors $\to$ field wires
- No hardcoded lattice geometry: the **same** ansatz code is correct for open or closed chains, 1D or 2D, because it is built from whatever bonds are actually present
- Circuit: prepare $|-\rangle^{\otimes N}$, then p layers of $\prod_{\langle i,j \rangle} e^{-i\gamma Z_i Z_j}$ followed by $\prod_i e^{-i\beta X_i}$ — 2 parameters per layer, not $O(N)$
- Respects the model's own $\mathbb{Z}_2$ symmetry by construction, which is a large part of why it avoids barren plateaus

Backup: gradient variance, precisely

- Sample n random parameter initializations; compute $\partial_\theta \langle H \rangle$ (exact, adjoint differentiation) at each
- Take the variance of each gradient **component** across the samples, then average over components (not one fixed angle — that was the pre-fix version of this exact plot)
- HEA only: exclude its leading angle — a leading R_z acting on $|0\rangle$, an unobservable global phase with identically zero gradient by construction, not a trainability signal. HVA's first angle is a genuine ZZ rotation (its **largest** gradient component) and is kept
- A variance that shrinks exponentially with N means: almost every random starting point sits on a flat plain — no local signal to climb down

Backup: why stop at N=16?

- Exact ground truth needs the full statevector: 2^N complex amplitudes — already 65,536 entries at $N = 16$
- Entanglement entropy needs an SVD of that state reshaped into a $2^{N/2} \times 2^{N/2}$ matrix at every field, every layout — the actual bottleneck, not the VQE circuits themselves
- The `qspin` dataset itself only ships `1x4`/`1x8`/`1x16` (chain) and up to `4x4` (rectangular) — there is no larger **exact** ground truth to compare against even if we wanted one
- VQE/QAOA circuits alone would happily scale further; it is the exact cross-checks (ED, entanglement, JW) that anchor N at 16

Backup: Jordan-Wigner, the diagonalization in full

- After Jordan-Wigner + Fourier transform ($\hat{a}_j = N^{-1/2} \sum_q \hat{a}_q e^{iqj}$), the Hamiltonian block-diagonalizes into 2×2 sectors $(\hat{a}_q, \hat{a}^\dagger_{-q})$: $\hat{H} = \sum_{q>0} \begin{pmatrix} \hat{a}^\dagger_q & \hat{a}_{-q} \end{pmatrix} \begin{pmatrix} -J\cos q - gJ & -iJ\sin q \\ iJ\sin q & J\cos q + gJ \end{pmatrix} \begin{pmatrix} \hat{a}_q \\ \hat{a}^\dagger_{-q} \end{pmatrix}$
- Diagonalizing this 2×2 kernel gives two bands $\lambda_\pm(q) = \pm J\sqrt{(\cos q + g)^2 + \sin^2 q} = \pm J\sqrt{1 + g^2 + 2g\cos q}$
- Ground state fills every negative-energy mode (λ_- band); the gap is $\Delta(q) = \lambda_+(q) - \lambda_-(q)$, minimized at $q = \pi$ (this q's origin is shifted by π relative to the main-deck k), vanishing there exactly at $g = 1$
- This is the same physics as the main-deck dispersion slide, carried through to the explicit matrix diagonalization

Backup: codebase layout

```
data/               cached qspin .h5 files (git-ignored)
src/
  core/
    loader.py             qml.data.load -> typed IsingData dataclass
    plotting.py           shared style / N->color map / save_fig
    parallel.py           process-pool map for multi-restart VQE grids
  physics/
    analysis.py           order parameter, Binder cumulant, data collapse
    entanglement.py       entropy, central charge (open + closed)
    finite_size_scaling.py   h_c(N) extrapolation, central charge vs N
    boundary_comparison.py   open vs closed h_c
    lattice_2d.py         rectangular-lattice analysis
    jordan_wigner.py      analytic free-fermion cross-check
    classical_shadow.py   shadow-based order-parameter reconstruction
  ansatz/
    vqe.py / vqe_hva.py           HEA / HVA+QAOA ansätze + sweeps
    optimizer_trajectory.py       per-step energy-error trajectories
    vqe_error_profile.py          dE(h) + gradient variance vs field
    trainability.py / benchmark.py   gradient variance, N x depth sweep
                          16 modules, 2151 lines
tests/              mirrors src/ (core/physics/ansatz), 63 tests, pytest
                    621 lines; plotting/parallel exercised indirectly
plots/              generated figures (PNG), referenced by both outputs below
report.md         written report — embeds plots/*.png
presentation/            these slides — embed the *same* plots/*.png
```

Backup: how a result becomes a slide

- `qspin` dataset (cached HDF5) $\rightarrow$ `loader.load_ising()` $\rightarrow$ one typed `IsingData` per (lattice, periodicity, layout)
- Each physics question is one `src/*.py` module: takes `IsingData` in, writes one PNG to `plots/` out, via the shared `plotting.py` helpers
- `report.md` and `presentation/tfim_talk.typ` both embed those same PNGs directly by relative path — no numbers are re-typed by hand, only re-read from the code's own printed output
- `pytest` (63 tests) gates every module before its plot is trusted; a code-quality pass fixed a stale duplicate and 5 dead imports, and excluding a trivial h=0 point moved the headline from 38x to 23x — which then **held** under a later multi-restart (median $\pm$ IQR) rerun, all caught/confirmed precisely **because** the pipeline is this direct

Backup: dataset provenance

- PennyLane's `qspin` collection: exact-diagonalization ground truth hosted by Xanadu, downloaded once and cached locally as HDF5
- Not our own ED code — deliberately independent of everything we compute, so it can act as ground truth rather than a second copy of our own bugs
- `shadow_basis`/`shadow_meas` are pre-simulated randomized single-qubit Pauli measurements of the **same** exact ground states — the shot noise is real, only the "device" is simulated
- https://pennylane.ai/datasets/transverse-field-ising-model